PCB Inspection Report

Timestamp	2026-04-05 15:39:16
PCB Name	Unnamed PCB
Source	upload
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Technical Specifications

Operating Voltage	3.3V-5V (Typical IoT)
Current Draw (Max)	< 500mA
Operating Temperature	-20°C to +85°C
Communication Protocol	I2C/SPI/UART/WiFi
Power Supply Type	DC Regulated
Compliance Standards	RoHS/CE/FCC

Detected Defects

#	Label	Confidence	Severity	Impact	BBox
1	Missing_hole	0.7954	High	Function incomplete	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	High	Function incomplete	[802, 55, 848, 100]
3	Missing_hole	0.6693	High	Function incomplete	[705, 55, 745, 98]
4	Missing_hole	0.649	High	Function incomplete	[532, 274, 553, 297]
5	Missing_hole	0.5945	High	Function incomplete	[108, 198, 131, 222]
6	Missing_hole	0.3384	High	Function incomplete	[598, 56, 639, 98]
7	Missing_hole	0.3038	High	Function incomplete	[954, 704, 973, 722]

Component Analysis

Microcontroller	ESP32/STM32/RP2040
Sensors	Temperature/Humidity/Motion
Communication	WiFi/Bluetooth/Zigbee
Power Management	LDO/Boost Converter
Connectors	USB/JST/Pin Headers
Test Points	TP1-TP8 (Power/GND/Signals)

Explanation

1) Inspection Status
- Overall status: DEFECTIVE
- Risk level: HIGH
- Predicted failure probability: 70.06%
2) Defect Assessment
- Defect 1: Missing_hole | model_conf=0.80 | fused_conf=0.80 | bbox=[1192, 280, 1214, 301]
- Defect 2: Missing_hole | model_conf=0.69 | fused_conf=0.69 | bbox=[802, 55, 848, 100]
- Defect 3: Missing_hole | model_conf=0.67 | fused_conf=0.67 | bbox=[705, 55, 745, 98]
- Defect 4: Missing_hole | model_conf=0.65 | fused_conf=0.65 | bbox=[532, 274, 553, 297]
- Defect 5: Missing_hole | model_conf=0.59 | fused_conf=0.59 | bbox=[108, 198, 131, 222]
- Defect 6: Missing_hole | model_conf=0.34 | fused_conf=0.34 | bbox=[598, 56, 639, 98]
- Defect 7: Missing_hole | model_conf=0.30 | fused_conf=0.30 | bbox=[954, 704, 973, 722]
3) Operational Risk Impact
- High probability of field failure if unreworked defects remain.
4) Recommended Actions
- Rework defects by label priority (short/open/missing first).
- Reinspect repaired regions under magnification and AOI.
- Perform continuity + isolation checks on affected paths.
5) Release Decision Guidance
- Hold production release until full rework and electrical validation are complete.
- Full rework includes repairing all detected defects with appropriate techniques (soldering, component replacement, trace repair).
- Electrical validation encompasses continuity testing, isolation checks, and functional testing under operational conditions.
- Release without validation risks field failures, warranty claims, and potential safety issues in critical applications.
- Validation timeline should include thermal cycling and stress testing to ensure long-term reliability.
- Gemini unavailable reason: GEMINI_API_KEY is missing or placeholder.
- This fallback explanation is generated without Gemini response.

Testing Procedures

1. Power Supply Test: Apply rated voltage and verify current draw < specified max
2. Continuity Check: Verify all power and ground nets using multimeter
3. Signal Integrity: Check clock signals, communication lines for proper levels
4. Functional Test: Load firmware and verify sensor readings/communication
5. Thermal Test: Monitor temperature rise under load (should be < 10°C above ambient)
6. Burn-in Test: 24-48 hour operation at elevated temperature for early failure detection

Repair Suggestions

- Missing_hole: Replace missing component and verify BOM compliance. Check for tombstoning or skew during reflow. Perform ICT test post-repair.

Manufacturing Notes

• Handle with ESD precautions - all components are ESD sensitive
• Store in moisture-proof packaging if not immediately assembled

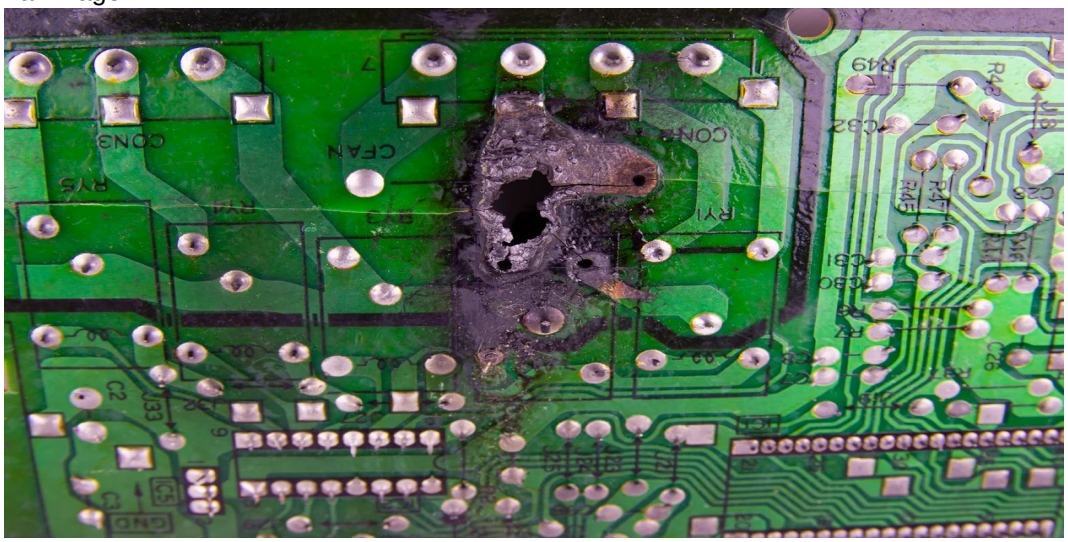
• Verify solder paste age and storage conditions before use
• Calibrate reflow oven profile for component types and board thickness
• Use nitrogen atmosphere for lead-free soldering if available
• Allow 24-hour cure time for conformal coating if applied

Quality Metrics

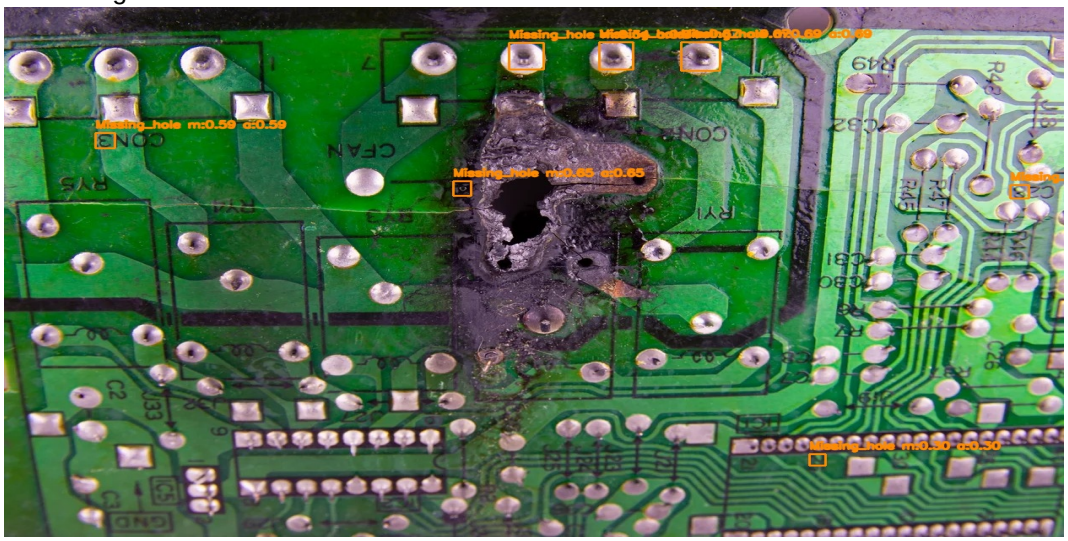
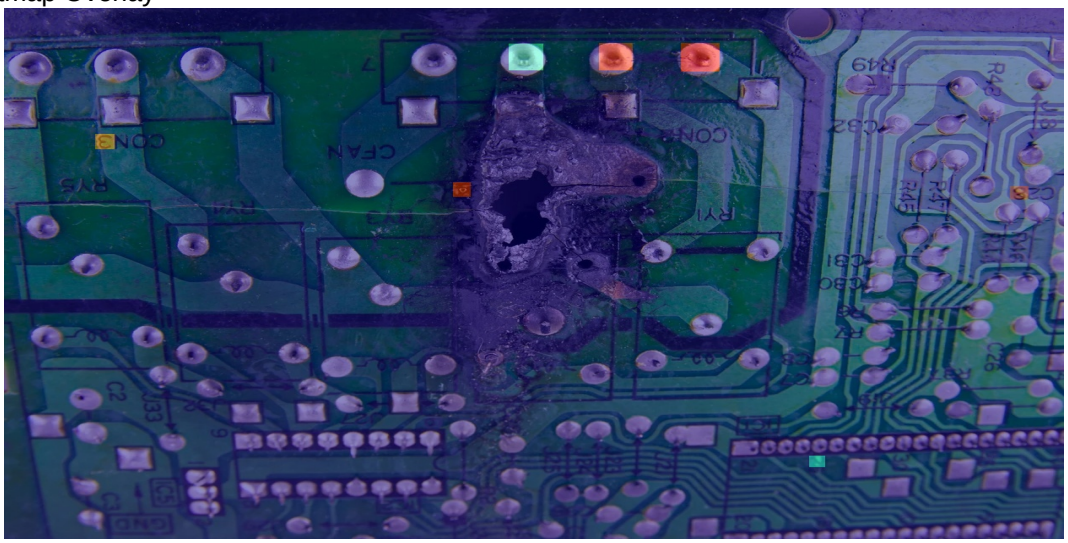
Defect Density	7 defects / 50-100 cm² (estimated) cm²
Process Yield	29.9% (estimated)
Inspection Coverage	Visual defects + AI analysis
Detection Threshold	95% confidence minimum
False Positive Rate	< 5% (typical)
MTBF Estimate	< 1,000 hours

Inspection Images

Original Image



Detection Image



Heatmap Overlay